

Job Application Automation

Comprehensive Implementation Plan

1. Overview

Automated pipeline that searches job platforms, matches your resume against job descriptions, tailors the resume when there's a mismatch, auto-applies, and tracks everything in Google Sheets + Gmail.

2. Tech Stack

Component	Choice
Language	Python 3.11+
Browser Automation	Playwright (headless)
LLM Providers	OpenAI (GPT-4o mini) + Anthropic (Claude 3.5 Sonnet)
Resume Parsing	pdfplumber (PDF) / python-docx (DOCX)
Tracking	gsread (Google Sheets) + Gmail API
Deployment	GitHub Actions (manual trigger)
Secrets	GitHub Secrets + local .env

3. Platform Targets

Platform	Priority	Approach
LinkedIn	Must	Playwright search + apply
Indeed	Must	Playwright search + apply
Wellfound (AngelList)	Must	Playwright search + apply
Naukri	Nice-to-have	Playwright search + apply
Greenhouse / Lever / Workday	Nice-to-have	Generic ATS form filler

4. Project Structure

```
job-apply-automation/  
+-- .github/workflows/run-automation.yml  
+-- src/  
|   +-- main.py           # Orchestrator  
|   +-- config.py        # Env vars, paths  
|   +-- resume/  
|       +-- parse.py      # PDF/DOCX/MD -> text  
|       +-- analyze.py    # LLM resume-JD matching  
|       +-- tailor.py     # LLM section rewrite  
|       +-- export.py     # Text -> PDF/DOCX  
|   +-- search/  
|       +-- base.py       # Abstract searcher  
|       +-- linkedin_search.py  
|       +-- indeed_search.py  
|       +-- wellfound_search.py
```

```

| | +-- naukri_search.py
| | +-- greenhouse_search.py
| | +-- lever_search.py
| | +-- workday_search.py
| +-- apply/
| | +-- base_applicator.py
| | +-- linkedin_apply.py
| | +-- indeed_apply.py
| | +-- wellfound_apply.py
| | +-- ats_apply.py
| +-- tracking/
|   +-- sheets.py           # Google Sheets CRUD
|   +-- email_monitor.py    # Gmail API
+-- data/resumes/
+-- requirements.txt
+-- README.md

```

5. Core Flow

Trigger (manual) >> Search all platforms >> For each job found:

- Scrape the full job description from the listing page
- Parse your resume (PDF/DOCX/MD -> plain text)
- LLM evaluates match score between resume and job description
- If score $\geq$ threshold: Proceed to apply with current resume
- If score $<$ threshold: LLM rewrites resume sections, injects ATS keywords, exports tailored PDF, then applies
- Log to Google Sheets: Company | Role | Date | Status | URL | Notes
- Gmail API watcher monitors inbox for replies -> updates Sheet status

6. LLM Strategy

Model	Role	Rationale
Claude 3.5 Sonnet	Job analysis & resume-JD matching	Long context, nuanced understanding
GPT-4o mini	Resume section rewriting	Structured output, cost-effective
Hybrid	Keyword injection for ATS optimization	LLM rewrites + targeted keyword insertion

7. Implementation Phases (11 Phases)

#	Phase	Deliverables
1	Foundation	config.py, parse.py, requirements.txt, CI workflow
2	LLM Client	Unified OpenAI + Anthropic wrapper
3	Resume Analyzer & Tailor	analyze.py, tailor.py, export.py
4	Job Search (core)	LinkedIn, Indeed, Wellfound searchers
5	Job Search (extended)	Naukri, Greenhouse, Lever, Workday
6	Auto-Apply (core)	LinkedIn, Indeed, Wellfound applicators
7	Auto-Apply (ATS)	Generic ATS form filler
8	Google Sheets Tracking	sheets.py - CRUD operations
9	Gmail Monitor	email_monitor.py - reply detection

10	Orchestrator	main.py - wires everything together
11	Deployment	GitHub Actions secrets + docs

8. GitHub Secrets Required

Secret	Purpose
OPENAI_API_KEY	GPT-4o mini
ANTHROPIC_API_KEY	Claude 3.5 Sonnet
GM_CREDENTIALS_JSON	Google service account (JSON)
GM_TOKEN_JSON	Gmail OAuth token (JSON)
LINKEDIN_USER	LinkedIn login email
LINKEDIN_PASS	LinkedIn password
INDEED_USER	Indeed login email
INDEED_PASS	Indeed password
WELLFOUND_USER	Wellfound login email
WELLFOUND_PASS	Wellfound password
RESUME_BASE64	Base64-encoded base resume PDF

9. What You Need to Prepare

- Base resume (Markdown recommended for easy LLM editing, then export to PDF/DOCX)
- Google Cloud Project with Sheets API + Gmail API enabled (service account + OAuth)
- Active accounts on LinkedIn, Indeed, and Wellfound
- OpenAI API key and Anthropic API key with billing enabled